Thermal Simulation of Processing-in-Memory Devices using HotSpot 7.0

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Abstract*—3D Integrated Circuits (3D ICs) offer many advantages over their 2D counterparts, such as increased bandwidth and energy efficiency, that has lead to the popularity of commercial products such as HBM and HMC. However, they also have well-known thermal challenges that need to be resolved before they can be integrated with high performance processors or Processing in Memory (PIM). Microfluidic cooling is a promising solution to these thermal challenges, but its associated design space is large enough and complex enough that we need a design space exploration tool. We extend HotSpot, an existing thermal simulator, to support flexible modeling and simulation of microfluidic cooling designs. We believe this tool will enable research into novel microfluidic cooling techniques, ultimately resulting in higher performance 3D ICs with reduced thermal constraints.

***Keywords*—In-memory computing, processing-in-memory, 3D-IC, 3D-memory, thermal modeling, microfluidic cooling, thermal management**

I. Introduction

3D Integrated Circuits (3D ICs) have grown in popularity in recent years due to their improved bandwidth and increased energy efficiency over their 2D counterparts. However, 3D ICs also tend to have higher heat fluxes than planar chips. Furthermore, it is difficult to remove the heat produced within a 3D IC using traditional cooling techniques (heat sinks, heat pipes, etc.), because heat produced within the stack has a greater thermal resistance to travel through before reaching the cooling solution. For example, both HBM [1] and HMC [2] follow a design pattern in which a stack of memory dice is placed on top of a logic die. Often, the cooling solution used for this design pattern is a heat sink placed on top of the entire stack. However, the logic die produces most of the heat within the stack, and the memory dice only increase the thermal resistance between the logic die and the heat sink. Nai et al. evaluated an HMC 1.1 prototype [3] and found that in order to achieve full bandwidth utilization, the HMC 1.1 prototype required an active heat sink. A passive heat sink would result in shutdown with junction temperatures reaching 90°C. They also simulated an HMC 2.0 stack at full bandwidth utilization and found that the temperature reached 81°C (dangerously close to the maximum desired temperature of 85°C) even with a commodity-server active heat sink.

It was also shown in [4] that if 8 DRAM dice are stacked on top of a logic die, even with a high-end server active heat sink, the logic die's power budget is limited to 55 watts if the stack temperature is to remain below 85°C. While this may be suitable for some applications, it suggests that more advanced cooling solutions will be necessary to create high-performance 3D ICs with more powerful processors that expect larger power budgets.

The power dissipation of the memory dice can also ex acerbate a 3D IC's thermal challenges [4]. For instance, the increased bandwidth available in 3D ICs makes them an enticing choice for incorporating Processing in Memory (PIM). However, not only can the PIM computations themselves introduce more heat [3], but many PIM solutions result in more memory cells being active at one time. For instance, Ambit [5] performs bulk AND, OR, and NOT operations in DRAM by activating multiple rows within a subarray simultaneously and by adding transistors to the sense amplifiers. The increased number of active memory cells and the inclusion of more transistors in the memory dice will increase the memory's power usage and, consequently, the heat that it dissipates.

The problem is that traditional cooling techniques tend to remove heat only at a surface, while in a 3D IC heat is being produced throughout a volume. Cooling 3D ICs effectively will thus require more advanced cooling techniques that can remove heat throughout a stack's volume. Many solutions have been suggested, including passive methods such as thermal vias, heat pipes, and thermosyphons, and active methods such as pumped loops and refrigerators [6]. However, we believe that microfluidic cooling strikes a good balance between effectiveness and practicality. However, there are many design choices when implementing microfluidic cooling, making a design space exploration tool necessary.

In Section II, we describe some background on microfluidic cooling and discuss the need for a design space exploration tool further. In Section III, we discuss HotSpot 6.0's thermal model and how we've extended it to support microfluidic design space exploration. In Section IV, we discuss some of the implementation details, especially regarding differential equation solvers. Finally, in Section V, we perform some example simulations to demonstrate HotSpot's utility.

II. Background: Microfluidic Cooling

Microfluidic cooling is a thermal management technique in which coolant is pumped through a 3D IC, usually through small cavities called microchannels [7]. Microfluidic cooling has been shown to be very effective, with Erp et al. reporting the ability to remove heat fluxes exceeding 1.7 kilowatts per square centimeter [8]. When using microchannels in a 3D IC, they can effectively remove heat throughout the volume of the stack, making them more effective than traditional surface-based cooling techniques.

While microfluidic cooling is very promising, it does come with a complex design space. Perhaps the most impactful decision in microfluidic cooling is choosing where exactly to place the channels. Microchannels can be placed in a heat sink or heat spreader, but we believe that for 3D ICs, they will be most effective when placed between layers. It has also been shown that the geometry of the microchannels can have significant effects on the both the effectiveness of

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the cooling and the efficiency of the pumping system [9]. Furthermore, there are closely related cooling techniques to consider, like pin fin cooling and pumping coolant through the micro-gaps present between layers, such as the inter-chip cooling proposed in DARPA's ICECool project [7].

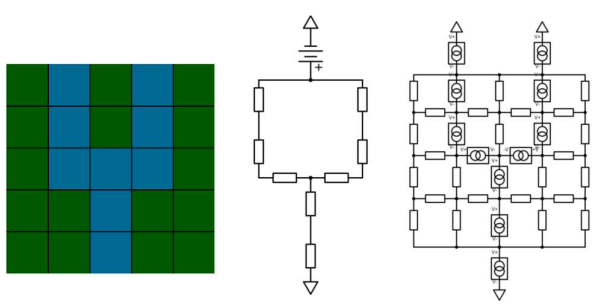
With such a large design space to explore, the research community needs a tool for modeling and simulation of microfluidic cooling. To be useful, this tool should be flexible enough to support many different microfluidic cooling techniques, including those already discussed and novel techniques yet to be discovered. It should also be employed early in the design process (pre-RTL) to enable researchers to quickly investigate new designs and hone in on the promising ones.

III. HotSpot's Thermal Mode

To create a tool that allows for thorough early-stage design space exploration, we start with HotSpot 6.0 [10], [11], an existing thermal simulator capable of simulating 3D ICs, and extend its thermal model to support microfluidic cooling. HotSpot requires very little information (such as a 3D IC's dimensions and power dissipation values) to perform a simulation, making it suitable for the pre-RTL design space exploration that we want to achieve.

HotSpot 6.0 models a 3D IC by discretizing it into an array of 3D cells, then modeling each cell as one node in a thermal circuit. This circuit can then be analyzed using well established circuit analysis techniques to find the temperatures at each node, which correspond to the temperatures within each cell throughout the 3D IC.

One of the limitations of HotSpot 6.0, however, is that it can only model heat transfer due to conduction. To support microfluidic cooling, we extend the thermal model to also support modeling heat transfer due to convection. In the following sections, we discuss how conductive heat transfer is modeled in HotSpot 6.0 and the extensions that we've made to allow modeling convective heat transfer. An example of a microfluidic cooling layer and the circuits that we use to model it are shown in Figure 1 [12].



(a) (b) (c)

Figure 1. (a) Example microfluidic circuit that is discretized into a 5x5 grid. Green cells are solid cells and blue cells are fluid cells. (b) Corresponding pressure circuit. (c) Corresponding thermal circuit.

A. Modeling Conductive Heat Transfer

Conductive heat transfer may occur between any two adjacent solid cells or any two adjacent fluid cells in the discretized 3D IC. To model this, we connect each node in the thermal circuit to all adjacent nodes of the same type (solid or fluid) through thermal resistances.

We model heat flow from the center of one cell to the center of the adjacent cell. Since adjacent cells may be different materials, we can think of the thermal resistance connecting the nodes as being two thermal resistances in series: one modeling the conductive heat transfer from the center of the first cell to the edge of the first cell, and one modeling the conductive heat transfer from the edge of the second cell to the center of the second cell. The value of the total thermal resistance is given by (1), in which t is the distance between the centers of the cells, k1 is the thermal conductivity of one cell's material, k2 is the thermal conductivity of the other cell's material, and A is the cross-sectional area of the interface between the cells.

$$R_{\text{th}} = \frac{t}{2k_1A+t} + \frac{t}{2k_2A} \tag{1}$$

For transient simulations, we also need to take into account each cell's thermal capacitance. We model this by connecting each node to ground through a capacitance whose value is given by (2), where C is the cell's volumetric heat capacity, t is the cell's thickness, and A is the cell's cross-sectional area.

$$C_{\text{th}} = CtA \tag{2}$$

B. Modeling Convective Heat Transfer

We adopt a model very similar to the 4RM-based Compact Transient Thermal Model (CTTM) used in 3D-ICE [13], [14] for convective heat transfer. This thermal model uses the fluid flow rate between every pair of fluid cells. However, finding these fluid flow rates is nontrivial since in HotSpot, we don't make any assumptions about the microchannel geometry. To find the flow rates, we model each microfluidic cooling layer using a pressure circuit by taking advantage of the similarities between Ohm's Law and the Hagen-Poiseuille Law [15]. In the pressure circuit, voltage is analogous to pressure, current flow is analogous to fluid flow, and electrical resistance is analogous to hydraulic resistance. In the following sections, we describe how we use the pressure circuit to find these flow rates and how we then use the flow rates to derive component values in the thermal circuit.

1) The Pressure Circuit: In order to derive the component values for the thermal circuit, we need to know the fluid flow rates between any given pair of fluid cells. To find these, we model each microfluidic cooling layer as a pressure circuit. The pump is modeled as a voltage source, and the hydraulic resistances of the channels are modeled as electrical resistances. The resistance between a pair of cells is given by (3), where h, w, and L are the height, width, and length of the cells, respectively, and η is the dynamic viscosity of the fluid [15].

$$R_{hyd} = \frac{12\eta L}{1-0.63(h/w)}\frac{1}{h^3 w} \tag{3}$$

After finding the hydraulic resistances between every pair of fluid cells, we perform node-voltage analysis to find the pressure in each fluid cell. Once we have found the pressure and resistance between every pair of fluid cells, we find the flow rates using a simple application of the Hagen-Poiseuille Law. Figure 1b shows a schematic of the pressure circuit for the microchannel given in Figure 1a.

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2) The Thermal Circuit: Convective heat transfer from a solid cell to an adjacent fluid cell is modeled using a thermal resistance between the two cells. We again assume heat flow between the centers of the cells, so the thermal resistance can be thought of as two resistances in series: one representing conductive heat transfer from the center of the solid cell to the edge of the solid cell, and one representing convective heat transfer from the edge of the fluid cell to the center of the fluid cell. The total thermal resistance is given by (4), where t is the distance between the centers of the cells, k is the thermal conductivity of the solid cell, h is the heat transfer coefficient, and A is the area of the interface between the cells.

$$R_{th} = \frac{t}{2kA} + \frac{1}{hA} \tag{4}$$

To model the heat flow between fluid cells due to the movement of the fluid, we connect adjacent fluid cells via current sources. A current source connecting one fluid cell to another fluid cell models heat transfer from the first fluid cell to the second. The values of the current sources is given by (5), where C is the volumetric heat capacity of the fluid, V is the volumetric flow rate of the fluid (which we found using the pressure circuit), and T_i is the temperature at the interface between the fluid cells.

$$J = CVT_i \tag{5}$$

In practice, we approximate T_i as the average of T_1 and T_2, where T_1 is the temperature of the cell from which the coolant is flowing and T_2 is the temperature of the cell to which the coolant is flowing: $T_i = (T_1 + T_2)/2$. Figure 1c shows a schematic of the thermal circuit for the microchannel shown in Figure 1a. For simplicity we have omitted the thermal capacitances, vertical thermal resistances to different layers, and the thermal resistances between fluid cells (adjacent fluid cells are connected via both thermal resistances and current sources).

IV. IMPLEMENTATION DETAILS

The thermal circuit based temperature calculation is a fast and straightforward way to predict the memory's temperature with different power consumption configurations. The simulation result of the logic layer concentrated in-memory computing model is described in figure 4. The temperature of the logic layer rises as power density increases. The increasing trends have different slopes due to the cooling features. The passive heat sink has the lowest cooling capacity of 4.0 K/W. The active heat sinks have more robust cooling capabilities of 2.0, 0.5, and 0.2 K/W for low-end, commodity-server, and high-end-server active coolers. The simulation results are validated using the simulation results from 3D-ICE [11], [12].

HotSpot 6.0 uses a 4th-order Runge Kutta (RK4) algorithm with adaptive step sizing as its transient differential equations solver. However, we've found that our changes to the thermal model have increased the time required for the RK4 solver to converge. In some larger simulations, this convergence time is prohibitive. As a result, we choose to implement a differential equations solver using the backward Euler method. We choose this method because it is the algorithm used in 3D-ICE [13], [14], which we use as a basis for our changes to HotSpot.

The choice of differential equations solver comes with a tradeoff. The RK4 method is more accurate, since it has a local truncation error of $O(h^5)$, while the backward Euler method has a local truncation error of $O(h^2)$. However, the backward Euler method is A-stable, while the RK4 method is not. We've found that the RK4 method provides a better balance of accuracy and simulation time for simulations that don't include microfluidic cooling, while the backward Euler method provides a better balance for simulations that include microfluidic cooling. The rest of this section describes the backward Euler solver and how we've implemented it in HotSpot in more detail. To use the backward Euler solver, we first represent the thermal circuit using (6).

$$GT + C\frac{dT}{dt} = P \tag{6}$$

For a circuit with n nodes, G is an $n \times n$ connectivity matrix that describes how all cells are connected via thermal resistances and current sources, C is an $n \times n$ diagonal matrix that contains the thermal capacitance of each cell, T is an $n \times 1$ vector of the temperatures of each cell, and P is an $n \times 1$ vector of the power dissipation of each cell. Before the simulation, G, C, and P are known, and we wish to find the temperatures T.

To solve (6) with the backward Euler method, we use (7), where h is the step size and f is a function representing the derivative dT/dt.

$$T_{n+1} = T_n + hf(t_{n+1}, T_{n+1}) \tag{7}$$

After solving (6) for dT/dt, inserting the result into (7), and rearranging to isolate T_{n+1}, we get (8).

$$\left(\frac{1}{h}C + G\right)T_{n+1} = \frac{1}{h}CT_n + P \tag{8}$$

Finally, we write (8) as the matrix equation $AT_{n+1} = B$ and solve it using the SuperLU matrix solving library [16], [17].

Solving the thermal circuit in steady-state simulations is much simpler. In steady-state simulations, we assume that all thermal capacitances have been fully charged and thus are effectively open circuits. This allows us to zero out the capacitance matrix C and solve the resulting equation with SuperLU.

V. SIMULATION RESULTS

The power budget limitation of in-3D-memory computing devices is investigated by using thermal modeling and simulation. The in-memory computing architecture can be modeled as a logic layer concentrated model or a distributed processing load model. As in-memory computing architecture dissipates more power in the memory layers than the conventional memory operation, the distributed model is appropriate to investigating thermal behaviors. The proposed modeling method is validated through comparison with a simulator 3D-ICE. We find the operating condition with different power consumptions in memory and logic layers, which sustained the operating temperature of the memory system below 85 °C.

To demonstrate HotSpot's utility as a design space exploration tool, we perform three sets of simulations. Our goal is both to show the value of exploring various microfluidic cooling configurations and to show HotSpot's ability to explore them. In our first simulation, we model a square die with a side length of 20.8 millimeters that is

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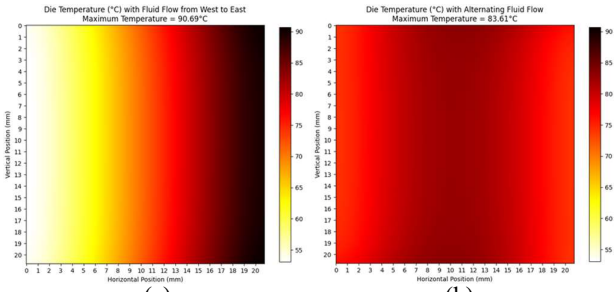
uniformly dissipating power with a density of 100 watts per square centimeter. In both simulations, this die is sitting atop a microfluidic cooling layer in which the microchannels run horizontally (from west to east) and span the width of the die. In the first simulation, coolant flows from west to east with a flow rate of 1 milliliter per minute. Figure 2a shows the resulting temperatures in the die. In the second simulation, every odd channel has coolant flowing from west to east and every even channel has coolant flowing from east to west such that every other channel has a reversed flow direction (still with a flow of 1 milliliter per minute). Figure 2b shows the resulting temperatures in the die. Not only does this simple change decrease the maximum temperature, but it also greatly decreases the temperature gradient across the chip, which can improve reliability [9].



(a) (b)

Figure 2. (a) Temperatures throughout a square die with uniform power dissipation and fluid flow from west to east. (b) Temperatures throughout the same die with uniform power dissipation and alternating fluid flow.

Next, we model a hypothetical processor architecture loosely based off of the Intel i7-3960X [18]. Figure 3 shows the floorplan. We also create a hypothetical power map based on the i7-3960X's published TDP of 130 watts. The die is still square with a side length of 20.8 millimeters, and we use the same two configurations for the microfluidic cooling layer, although we reduce the flow rate to 0.5 milliliters per minute because the die's power density is significantly lower than in the previous simulations. Figure 4a shows a heat map of the die with coolant flowing from west to east and Figure 4b shows a heat map of the die with every other channel's flow direction reversed. Again, it is apparent that not only has this simple change decreased the maximum temperature, but it has also decreased the temperature gradient across the die.

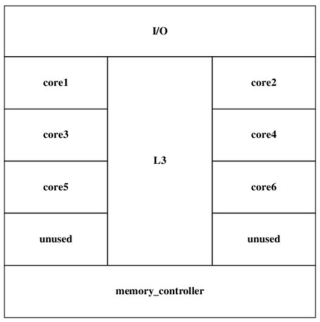


Figure 3. Floorplan of simulated die based on i7 processor

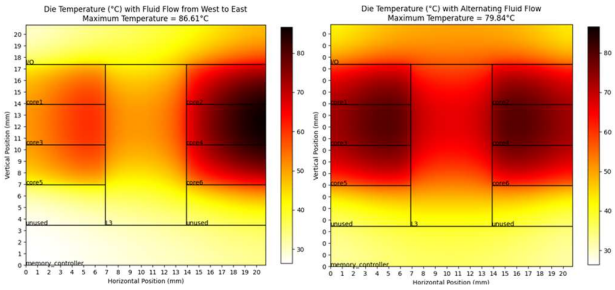
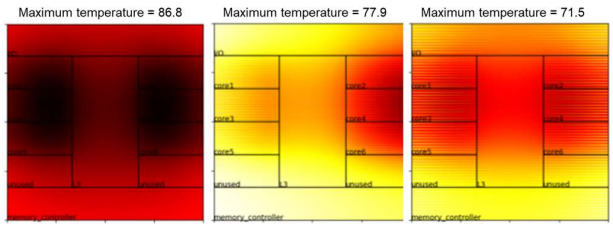


Figure 4. Temperatures throughout the die with non-uniform power dissipation and (a) unidirectional fluid flow from west to east and (b) alternating fluid flow.

Lastly, we perform thermal simulations of three different cooling configurations for a 3D stack with DRAM layers sitting on top of a logic layer to show the effect of different cooling configurations on a 3D-stacked PIM architecture. In the first cooling configuration, shown in Figure 5(a), we simulate cooling the PIM stack with a copper heat sink sitting on top of the stack. The resulting maximum temperature, 86.8°C, is above the maximum desired temperature of 85°C. In the second cooling configuration, shown in Figure 5(b), we cool the PIM stack via a microfluidic cooling layer sitting directly beneath the logic layer. The cooling layer has 138 microchannels that span the chip from left to right, each 50 m wide, with fluid flowing from left to right at a rate of 0.5 mL/min. This cooling configuration is clearly much more effective, reducing the maximum temperature by almost 9°C. However, notice that this cooling configuration results in a significant temperature gradient across the chip. This is due to the decreasing effectiveness of the coolant as it flows across the chip and warms. In the third and final cooling configuration, shown in Figure 5(c), we alternate the flow direction between the channels, so that one channel has fluid flowing from left to right, the next channel has fluid flowing from right to left, and so on. This not only decreases the maximum temperature of the chip by over 6°C, but it also spreads the heat out much more evenly across the chip.



(a) (b) (c)

Figure 5. Heat Map of PIM stack cooled by (a) copper heat sink on top of stack, (b) microfluidic cooling layer beneath heat source. All fluid flows from left to right, and (c) microfluidic cooling layer beneath heat source. The direction of fluid flow alternates between channels.

There are also many other cooling options we could explore. We could make changes to the microchannel geometry to only have channels underneath the cores, to flow from north to south, or to have a more unusual geometry. We could change the size of the channels as well as change the materials that make up the channel walls to have a higher conductivity. We may want multiple microfluidic cooling

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layers with different geometries that take Through Silicon Vias (TSVs) into account if this die were placed in a 3D IC. Different dies with different floorplans in the 3D IC may also benefit from different microfluidic cooling configurations. Our extensions to HotSpot's thermal model allow for simulating all of these cooling options with little effort.

VI. CONCLUSION

3D ICs are a very promising technology for the future. They offer ways to extend Moore's Law into the third dimension, improved energy efficiency, and reduced transmission losses from shortened interconnects. 3D ICs also offer increased bandwidth due to the ability to place TSVs over a layer's surface instead of being limited to pins located at the periphery. However, their increased heat flux requires cooling solutions to remove heat throughout their volume instead of merely at the surface Microfluidic cooling offers the ability to cool 3D ICs effectively, but it comes with a massive design space that needs to be explored. To support this design space exploration, we have extended HotSpot's thermal model to support simulating microfluidic cooling. This will give the research community access to a powerful new tool that allows simulations very early in the design process while being flexible enough to support many cooling designs.

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